

Polyanalytic Toeplitz operators: the linear closure is proper, but the generated C^* -algebra is all of $\mathcal{C}_1$

Full answer to Question 2 of arXiv:2308.11292

13 August 2026

Status. For every true polyanalytic level $k \geq 2$, both clauses of the source question have complete answers. The norm closure of the individual Toeplitz operators with bounded uniformly continuous symbols is a proper subspace of $\mathcal{C}_1$; indeed, every Weyl operator whose frequency lies on a Laguerre zero circle has distance exactly 1 from even the larger L^∞ -symbol range. In contrast, the C^* -algebra generated by those Toeplitz operators is exactly $\mathcal{C}_1$.

Source question

Let $F_{(k)}^2$ be the true k -th polyanalytic Fock space and

$$\alpha_z(A) = W_z A W_{-z}, \quad \mathcal{C}_1(F_{(k)}^2) = \{A : \|\alpha_z(A) - A\| \rightarrow 0 \ (z \rightarrow 0)\}.$$

The source asks whether

$$\overline{\{T_{f,(k)} : f \in \text{BUC}(\mathbb{C})\}}^{\|\cdot\|} = \mathcal{C}_1(F_{(k)}^2)$$

for $k \geq 2$, and asks the same about the C^* -algebra generated by these operators [1, p. 36, Question 2].

but it is not quite clear to us if this is already enough to answer Question 1 affirmatively in case $k = 2$.

Besides the characterizations of $\mathcal{C}_1$ discussed in Section 4, further characterizations are available for $k = 1$. It is known that $\mathcal{C}_1(F_{(1)}^2)$ equals $\overline{\{T_{f,(1)} : f \in \text{BUC}(\mathbb{C})\}}$ as well as $C^*(\{T_{f,(1)} : f \in \text{BUC}(\mathbb{C})\})$. Let us again restrict our discussion to the case $k = 2$. By Corollary 3.10, the Weyl operators W_z with $|z| = 1$ are not contained in $\{T_{f,(2)} : f \in L^\infty(\mathbb{C})\}$. It is not clear if they are contained in the uniform closure of the set. By Proposition 3.8, all Weyl operators are at least contained in the weak* closure.

Nevertheless, using that most Weyl operators can be written as Toeplitz operators, cf. Remark 6, $\{T_{f,(2)} T_{g,(2)} : f, g \in \text{BUC}(\mathbb{C})\}$ is easily seen to contain all Weyl operators. Therefore, there is at least no obvious obstacle preventing $C^*(\{T_{f,(2)} : f \in \text{BUC}(\mathbb{C})\})$ from being all of $\mathcal{C}_1(F_{(2)}^2)$. But the problem remains open:

Question 2. Does $\overline{\{T_{f,(k)} : f \in \text{BUC}(\mathbb{C})\}}$ equal $\mathcal{C}_1(F_{(k)}^2)$ for $k \geq 2$? How about $C^*(\{T_{f,(k)} : f \in \text{BUC}(\mathbb{C})\})$?

36

Source evidence: arXiv:2308.11292, PDF page 36.

Answer

Put

$$\mathfrak{L}_k = \overline{\{T_{f,(k)} : f \in \text{BUC}(\mathbb{C})\}}^{\|\cdot\|}, \quad \mathfrak{A}_k = C^*\{T_{f,(k)} : f \in \text{BUC}(\mathbb{C})\}.$$

Theorem 1. For every $k \geq 2$,

$$\mathfrak{L}_k \subsetneq \mathcal{C}_1(F_{(k)}^2) \quad \text{and} \quad \mathfrak{A}_k = \mathcal{C}_1(F_{(k)}^2).$$

More precisely, if

$$Z_k = \{\rho \in \mathbb{C} : L_{k-1}^0(|\rho|^2) = 0\},$$

then, for each $\rho \in Z_k$,

$$\text{dist}(W_{-\rho}, \{T_{f,(k)} : f \in L^\infty(\mathbb{C})\}) = 1.$$

Proof intuition. Toeplitz quantization at level k is a Fourier multiplier whose multiplier is a Gaussian times L_{k-1}^0 . It erases exactly the finitely many frequency circles Z_k . A contractive Fourier–Følner average isolates a single erased Weyl frequency, proving that it cannot even be approximated in norm by individual Toeplitz operators. Multiplication changes the picture: any Weyl operator is a product of two Toeplitz Weyl operators at non-erased frequencies. Those generated Weyl operators can shift a compact operator spectrum away from Z_k ; there the level- k multiplier is locally invertible. Compact-spectrum approximation then recovers every shift-continuous operator.

Fourier–Weyl setup

Write

$$S_k = k_{0,(k)} \otimes k_{0,(k)}, \quad Q_k(f) = S_k * f.$$

Up to the common harmless scalar used in the source, $Q_k(f) = T_{f,(k)}$. The source proves [1, Proposition 3.2 and (3.7)]

$$m_k(\xi) := \mathcal{F}_W(S_k)(\xi) = e^{-|\xi|^2/2} L_{k-1}^0(|\xi|^2). \quad (1)$$

Thus $Z_k = m_k^{-1}(0)$ is a finite nonempty union of concentric circles for $k \geq 2$. For

$$\gamma_\xi(w) = e^{i\sigma(\xi,w)},$$

the convolution theorem gives [1, Remark 6]

$$Q_k(\gamma_\xi) = c_k(\xi) W_{-\xi}, \quad c_k(\xi) \neq 0 \iff \xi \notin Z_k. \quad (2)$$

Both $f \mapsto Q_k(f)$ and the shift action are weak-* continuous, and

$$\alpha_z(Q_k(f)) = Q_k(\alpha_z f), \quad (\alpha_z f)(w) = f(w - z). \quad (3)$$

The sharp obstruction to linear norm density

Fix $\rho \in Z_k$. For $R > 0$, define

$$P_R(A) = \frac{1}{|B_R|} \int_{B_R} e^{i\sigma(\rho,z)} \alpha_z(A) dz.$$

For $A \in \mathcal{C}_1$ this is a Bochner integral, and $\|P_R(A)\| \leq \|A\|$. The Weyl relations give

$$\alpha_z(W_{-\rho}) = e^{-i\sigma(\rho,z)} W_{-\rho}, \quad P_R(W_{-\rho}) = W_{-\rho}.$$

For $f \in L^\infty(\mathbb{C})$, covariance (3) gives

$$P_R(Q_k(f)) = Q_k(g_R), \quad g_R = \frac{1}{|B_R|} \int_{B_R} e^{i\sigma(\rho,z)} \alpha_z(f) dz.$$

The net (g_R) is bounded in L^∞ . If a subnet converges weak-* to g , the Følner property

$$\frac{|(B_R + y) \triangle B_R|}{|B_R|} \longrightarrow 0$$

implies

$$\alpha_y(g) = e^{-i\sigma(\rho,y)} g \quad (y \in \mathbb{C}).$$

Consequently $g\overline{\gamma_\rho}$ is translation invariant, hence a.e. constant, so $g = c\gamma_\rho$. But $\rho \in Z_k$, and therefore (2) gives $Q_k(g) = 0$. Weak-* continuity of Q_k now shows that

$$P_R(Q_k(f)) \longrightarrow 0 \quad \text{ultraweakly.}$$

Contractivity of P_R and lower semicontinuity of the operator norm under ultraweak convergence yield

$$\begin{aligned} \|W_{-\rho} - Q_k(f)\| &\geq \|P_R(W_{-\rho} - Q_k(f))\| \\ &= \|W_{-\rho} - P_R(Q_k(f))\|, \\ \|W_{-\rho} - Q_k(f)\| &\geq \liminf_{R \rightarrow \infty} \|W_{-\rho} - P_R(Q_k(f))\| \geq \|W_{-\rho}\| = 1. \end{aligned}$$

The choice $f = 0$ gives the reverse inequality. The distance is exactly 1, and the first assertion of Theorem 1 follows.

Why multiplication recovers all of $\mathcal{C}_1$

Lemma 1. *Every Weyl operator belongs to $\mathfrak{A}_k$.*

Proof. Equation (2) gives $W_{-\xi} \in \mathfrak{A}_k$ whenever $\xi \notin Z_k$. Given arbitrary η , choose

$$a \notin Z_k \cup (\eta - Z_k).$$

Then $a, \eta - a \notin Z_k$. Hence W_{-a} and $W_{-(\eta-a)}$ belong to $\mathfrak{A}_k$, while their product is a unimodular scalar times $W_{-\eta}$. $\square$

We use three standard facts about the Arveson spectrum $\text{sp}_\alpha(A)$ of the norm-continuous action α :

$$\begin{aligned} \text{sp}_\alpha(h * A) &\subseteq \text{supp } \widehat{h}, \\ \widehat{\psi} = 1 \text{ near } \text{sp}_\alpha(A) &\implies \psi * A = A, \\ \text{sp}_\alpha(W_\eta A) &= \text{sp}_\alpha(A) + \varepsilon\eta \end{aligned} \tag{4}$$

for the fixed convention-dependent sign $\varepsilon \in \{1, -1\}$. These follow directly by taking the Fourier transform of the orbit $z \mapsto \alpha_z(A)$.

Lemma 2. *If $B \in \mathcal{C}_1$ has compact Arveson spectrum K with $K \cap Z_k = \emptyset$, then $B \in \mathfrak{L}_k$.*

Proof. Transport the irreducible Weyl representation on $F_{(k)}^2$ unitarily to the analytic Fock representation. Under this identification Q_k has multiplier m_k , whereas analytic Toeplitz quantization Q_1 has the nowhere-zero multiplier

$$m_1(\xi) = e^{-|\xi|^2/2}.$$

The analytic correspondence theorem says

$$\mathcal{C}_1 = \overline{Q_1(\text{BUC}(\mathbb{C}))}^{\|\cdot\|} \tag{5}$$

[2, Theorem 3.1]; the source also recalls this fact immediately before Question 2.

Choose $\psi \in \mathcal{S}(\mathbb{C})$ such that

$$\widehat{\psi} = 1 \text{ near } K, \quad \text{supp } \widehat{\psi} \cap Z_k = \emptyset.$$

Then $\psi * B = B$ by (4). From (5), choose $b_j \in \text{BUC}$ with $Q_1(b_j) \rightarrow B$. Hence

$$Q_1(\psi * b_j) = \psi * Q_1(b_j) \longrightarrow B.$$

Define $\theta \in \mathcal{S}(\mathbb{C})$ by

$$\widehat{\theta}(\xi) = \frac{m_1(\xi)}{m_k(\xi)} \widehat{\psi}(\xi).$$

The quotient is smooth on $\text{supp } \widehat{\psi}$, so this is legitimate. Put $f_j = \theta * b_j \in \text{BUC}$. In tempered Fourier–Weyl distributions,

$$\begin{aligned} \mathcal{F}_W(Q_k(f_j)) &= m_k \widehat{\theta} \widehat{b_j} \\ &= m_1 \widehat{\psi} \widehat{b_j} = \mathcal{F}_W(Q_1(\psi * b_j)). \end{aligned}$$

Injectivity of the tempered Fourier–Weyl transform gives

$$Q_k(f_j) = Q_1(\psi * b_j) \longrightarrow B.$$

Thus $B \in \mathfrak{L}_k$. $\square$

Completion of the proof of Theorem 1. Every $Q_k(f)$ with $f \in \text{BUC}$ is shift-continuous, and $\mathcal{C}_1$ is a C^* -algebra; therefore $\mathfrak{A}_k \subseteq \mathcal{C}_1$.

Conversely, let $A \in \mathcal{C}_1$. Choose an L^1 approximate identity (h_n) whose symplectic Fourier transforms have compact support. Norm continuity of the orbit gives

$$A_n = h_n * A \longrightarrow A, \quad K_n := \text{sp}_\alpha(A_n) \subseteq \text{supp } \widehat{h_n}.$$

The set Z_k is bounded. Choose η_n and the sign in (4) so that

$$(K_n + \varepsilon \eta_n) \cap Z_k = \emptyset.$$

Then $B_n = W_{\eta_n} A_n$ has compact spectrum disjoint from Z_k . Lemma 2 gives $B_n \in \mathfrak{L}_k \subseteq \mathfrak{A}_k$, while Lemma 1 gives $W_{\eta_n} \in \mathfrak{A}_k$. Hence

$$A_n = W_{\eta_n}^* B_n \in \mathfrak{A}_k.$$

Taking the norm limit proves $A \in \mathfrak{A}_k$, so $\mathfrak{A}_k = \mathcal{C}_1$. □

Scope and provenance

This settles every clause of the source's Question 2 for every $k \geq 2$. It does not claim an answer to the separate compactness-modulo-Berezin-kernel Question 1. Exact arXiv-id, title, quotation, author, and core-keyword searches through 13 August 2026, together with a direct-citation scan of the published source, found no later arXiv theorem settling Question 2. The proof above is a new result from this run. Its external inputs are the source's explicit multiplier, character, covariance, and tempered Fourier–Weyl identities, plus the analytic correspondence theorem (5).

References

- [1] R. Fulsche and R. Hagger, *Quantum harmonic analysis for polyanalytic Fock spaces*, arXiv:2308.11292 (2023; revised 2024). See Proposition 3.2, Remark 6, and Question 2 on PDF page 36.
- [2] R. Fulsche, *Correspondence theory on p -Fock spaces with applications to Toeplitz algebras*, arXiv:1911.12668; J. Funct. Anal. 279 (2020), 108661.